

Bias against certain high frequency DNA sequences at close proximity

Please allow me a few minutes to present a mystery regarding repetitive DNA sequences, of importance in understanding the physiology of cyanobacteria (major contributors to global photosynthesis). You'll see that my evidence, though convincing I think, is purely graphical. A statistical treatment would enable a more detailed analysis and broader presentation of the evidence. In the end, I'm going to ask for some advice, but to tell the truth, snagging you as a true collaborator would be much more appealing, if I'm able to elicit your interest in the problem.

Mystery: Apparent repulsion of highly repetitive interspersed sequences

Genomes to the uneducated eye may seem like a relentless series of random nucleotides – A, C, G, and T. They're not random, of course. For one thing they contain genes, the information to encode all our proteins. However, even without the ability to pick out genes from the seeming chaos, it is possible for us to discern clear patterns in the form of repetitive sequences, a feature of the genomes of almost all cellular organisms.

Repeated sequences can be divided amongst three general categories: Tandem repeats, long dispersed repeats, and short dispersed repeats (**Fig. 1**). Tandem repeats are best known to the public as the cause of several human genetic disorders. For example, Huntington's Disease results from an expansion of (CAG)_n repeats to a threshold length. About half of the human genome consists of long dispersed repeats, many of which are the remnants of ancient viral infections. The mechanisms by which tandem repeats and long dispersed repeats arise are fairly well understood.

In contrast, short dispersed repeats have a much lower profile. They are present as DNA sequences typically 6 to 9 nucleotides in length that are present in far greater abundance than predicted. Their origins and function within cells remain unknown, although compelling ideas have been put forward.¹

The most dramatic known instances of repeats of this type are HIP sequences (Highly Iterative Palindromes), found in most cyanobacteria. One version, HIP-1 (GCGATCGC)* occurs in cyanobacterial genomes at a frequency as much as 100 times what one would expect on the basis of the component mononucleotide frequencies. This overabundance is the highest known deviation from expectation of any sequence in any bacterial genome.

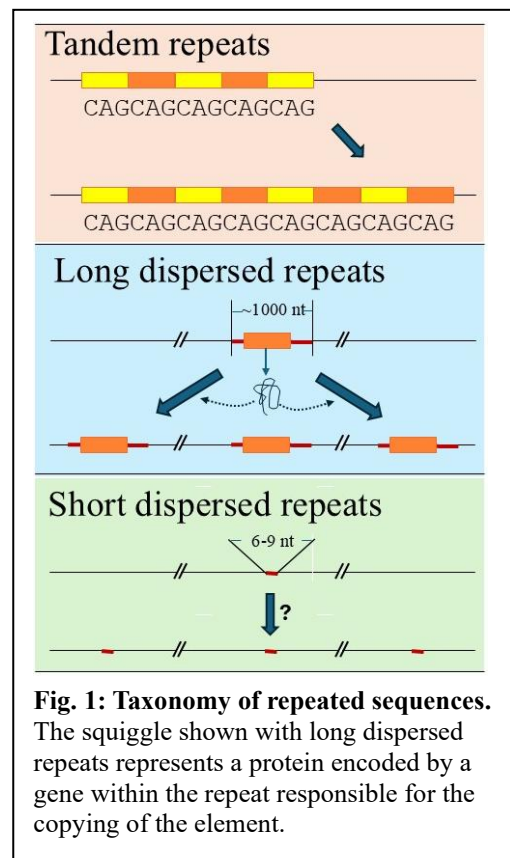


Fig. 1: Taxonomy of repeated sequences. The squiggle shown with long dispersed repeats represents a protein encoded by a gene within the repeat responsible for the copying of the element.

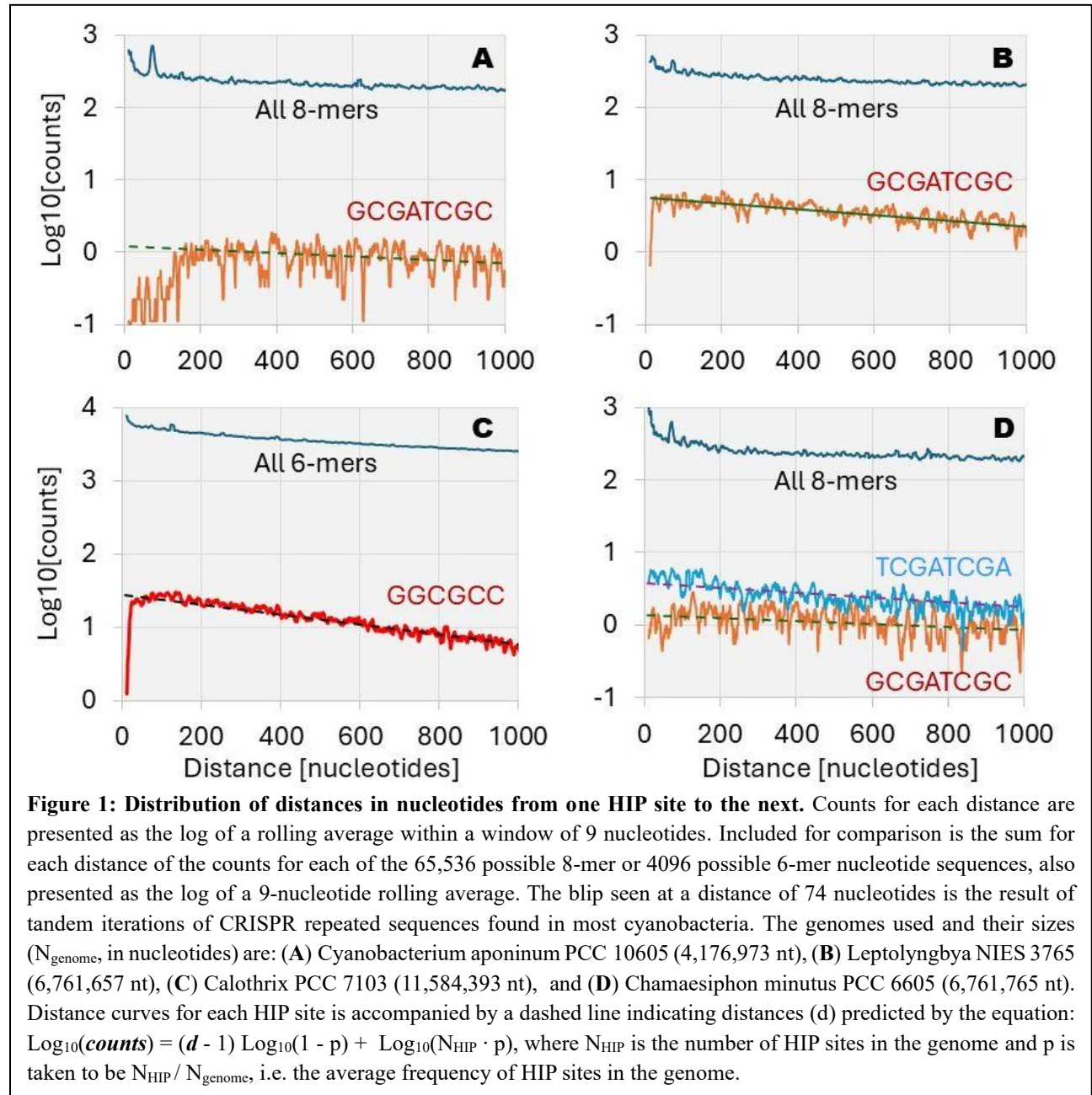
* GCGATCGC is considered a palindrome because DNA is double-stranded. Following nucleotide pairing rules (G pairs with C and A with T), a DNA palindrome is defined as a sequence where one strand reads the same as the other, but in the opposite direction. So GCGATCGC is palindromic because $\begin{matrix} \text{GCGATCGC} \\ \text{CGCTAGCG} \end{matrix}$ is a palindrome in the sense described.

I wondered whether HIP-1 sequences were randomly distributed in the genomes in which they occurred. If so, then one would expect that the distances between them to follow:

$$\text{Frequency of distance}(d) = p (1-p)^{d-1}$$

$$\text{HIP count}(d) = N_{\text{HIP}} p (1-p)^{d-1}$$

Where p is the average frequency of the sequence, taken to be (total HIP count) / (genome size). By and large, I found that this expectation was met in all cyanobacteria that had HIP-1 (which is the large majority of cyanobacteria). But the size distribution matched expectation only at distances above a certain threshold. You can see this in **Fig. 2**.



Almost all of the graphs of distance distributions in cyanobacteria fall into two major groups, typified by the graphs shown in Fig. 2A and Fig. 2B. In the first group (with distributions similar to Fig. 2A), there is strong deviation of distances from the predicted line, with underrepresentation of the HIP sequence that are less than 100 to 160 nucleotides distant. In the second group (with distributions similar to Fig. 2B), underrepresentation of the HIP sequence is strong but only with distances below about 20 nucleotides. Rare cyanobacteria do not have a highly abundant HIP-1 (GCGATCGC) sequence but have instead a different HIP sequence, argued to serve the same function within the bacterial cell.¹ Even in those cases, the HIP sequence is underrepresented at short distances (see Fig. 2C). In other rare cyanobacteria, HIP-1 sequences are abundant but less so than usual, and a different related HIP sequence is of high abundance. It is difficult to say whether there is underrepresentation in these cases, because the low numbers make visual interpretation difficult. Fig. 2D shows an example where there does not seem to be underrepresentation.

The evident underrepresentation of short distances may provide a clue as to the function of HIP sequences, a matter of controversy.^{1,2,3,4} Whether they participate in the repair of mutated DNA,¹ combining two sister molecules of DNA,^{2,4} or the overall physical structure of DNA,³ the sites clearly could inform our understanding of how cyanobacteria work. Presumably the general overrepresentation of HIP sequences is the result of selection for their function. If this function doesn't operate at short distances, as suggested by the deviations from random expectation seen in Figs 2A-2C, that would provide evidence that may distinguish amongst the various hypotheses regarding HIP function.

It is therefore a matter of some importance to establish whether the deviations are a general feature of HIP sequences. I've considered a few dozen cyanobacterial genomes, but it would be impractical to offer that many graphs as evidence. Furthermore, some might dispute the significance of the deviations. The solution would seem to be a statistical measure that could be applied to the data that would permit a compact presentation of all cyanobacterial genomes, the significance of the deviation (if present), and the distance at which counts no longer deviate significantly from expectation.

The only tool I have in my limited statistical toolbox to address the problem is a chi squared test, comparing the count and expected count (not log counts) of an HIP at a specific distance. I understand that it is not legitimate to use the test with averaged values, but without averaging, the HIP counts are very noisy. Is it possible to combine chi-squared values (from counts over a certain range) and obtain a p-value from their sum? Or perhaps there are wholly different ways, better methods I know nothing about, to consider the problem statistically.

References

1. Elhai J (2015). Highly iterated palindromic sequences (HIPs) and their relationship to DNA methyltransferases. [Life 5:921-948](#).
2. Akiyama H, Kanai S, Hirano M, Miyasaka H (1998). A novel plasmid recombination mechanism of the marine cyanobacterium *Synechococcus* sp. PCC 7002. [DNA Res 5:327-334](#),
3. Xu M, Lawrence JG, Durand D (2018). Selection, periodicity and potential function for Highly Iterative Palindrome-1 (HIP1) in cyanobacterial genomes. [Nucleic Acids Res 46:2265-2278](#).
4. Kamoku C, Nielsen DR (2024). Highly Iterated Palindrome 1 sequence improves *Synechococcus* sp. PCC 7002 transformation efficiencies in a homology- and methylation-dependent manner. [Biotechnol Prog 41:e3518](#).